

ENGR 422 – Computer Vision Assignment

Assignment I

Mohammed Mahdi Qasim

mm 22093

Classical Feature Detection for Medical Image Analysis

1. Introduction & Medical Motivation

Medical eye examination is considered a very important test that is a routine check-up and can be done either invasively or non-invasively. The non-invasive method depends most of the time on the Retinal fundus imaging, which is a method that works by screening ocular pathologies like Age-Related Macular Degeneration (ARMD) and Diabetic Retinopathy (DR). Observing and analyzing the complex vascular networks and finding anomalies (drusen, black spots, microaneurysms) can be a hard task even for specialists because of the background noise, variations in patients' cases, and camera lighting.



Figure 1: Non-processed Retinal Fundus

This is an experiment that applies classical computer vision detectors to several and different retinal images from real clinical data. The main detecting methods was evaluated in this experiment are edge detection for blood vessels, corner detection for vessel intersections, and blobs for bright lesions. Testing the limitations and strengths of the used algorithms, such as Canny, SIFT, Harris, and others, after pre-processing the images to make it ready to do the test in clinical practice environments. To add data and cover different cases, the test was applied on three main cases for the retinal fundus, normal, ARMD, and diabetic retinopathy.

2. Data

As mentioned, to have a comprehensive analysis and cover more scenarios, three different categories were used (normal, ARMD, and diabetic retinopathy), and these were collected from different dataset sources on Kaggle:

1- The dataset for ARMD images:

Mujib, R. (2023, May 11). *ARMD curated dataset 2023*. Kaggle.
<https://www.kaggle.com/datasets/rakhshandamujib/armd-curated-dataset-2023>

2- The dataset for normal images:

Linchundan. (2019, June 18). *1000 fundus images with 39 categories*. Kaggle.
<https://www.kaggle.com/datasets/linchundan/fundusimage1000>

3- The dataset for Diabetic retinopathy:

Soleimany, D. (2025, September 17). *Diabetic retinopathy image dataset*. Kaggle.
<https://www.kaggle.com/datasets/dankok/diabetic-retinopathy-image-dataset>

all these datasets used (RGB) images and had various images from the same category.

The testing was applied to 15 images, 5 images from each category, to ensure coverage of the data and find patterns and differences between the normal and sick, and even between the sick if it was ARMD or diabetic, with a language a computer can understand. For every 5 images, it was divided into easy, moderate, and highly difficult contrast scenarios.

3. Methods

3.1 Pre-processing

In most cases, when working with feature detection, the images are not perfectly ready to be implemented. Images may suffer from severe non-uniform illumination, quality, and poor background contrast, which leads to the urgent need for preprocessing methods, and for this test, two methods were applied:

1- Channel Extraction: in (RGB) retinal fundus images, we are dealing with blood vessels and these absorb green light strongly, so by taking in only the green channel from the RGB configuration for processing, the dark red vessels reach the best contrast with the bright background.

2- Contrast Enhancement and Denoising: since the images can be very messy due to the small and hard-to-see microscopic blood vessels, the need to dividing the image into small chunks and then apply the enhancement locally on these chunks with specific limitations of what to consider lead to the use of (CLASH) algorithm with the parameter (tileGridSize=[8,8]) as the size of each

chunk will be enhanced and (clipLimit=2.0) for the limit. In addition, a Gaussian Blur was also used for denoising with ksize(5,5) for all edge detection steps to deal with camera sensor noise.

3.2 Feature Detectors & Parameter Choices

3.2.1 Edge Detection

For this step, three methods were used to compare them, Canny against Sobel and Laplacian of Gaussian (LoG). The Canny method was applied with three different thresholds: Sensitive (20/50) for the noise detection, Balanced (40/100) for the moderate vessel detection, and Strict (80/200) to challenge detecting thick big vessels.

3.2.2 Corner Detection

In this detection, another comparison was done between Shi-Tomasi and Harris to detect vessel bifurcation. The Harris method was set with the parameters (blockSize=2), which set the size of the window that goes on the image to be 2x2 pixels, and the constant parameter (k=0.04). For Shi-Tomasi, the parameters were (maxCorners=1000) for the max number of points allowed on the image before stopping, and (qualityLevel=0.01) that scans all the 1000 corners with a sense of certainty of 0.01 to keep all the corners that are less than 1%.

3.2.3 Blob Detection

For the Blob detection, SIFT was used since it has the method of Difference of Gaussians (DoG) within it, and it was evaluated side by side with ORB, which is basically designed on the same principles as SIFT and rotated BRIEF, but the main purpose of it is to give faster results. They were both meant to locate blobs such as exudates.

4. Results & Analytical Requirements

4.1 Qualitative Observations

4.1.1 Preprocessing

It was clear to the normal human eye that for all the classes that were tested (normal, ARMD, Diabetic retinopathy), the green + CLASH filter gave the best results for making the blood vessels more visible and much easier to detect compared to the original, standard grayscale, or even the green channel without CLASH.



Figure 2: Preprocessing for the normal class.

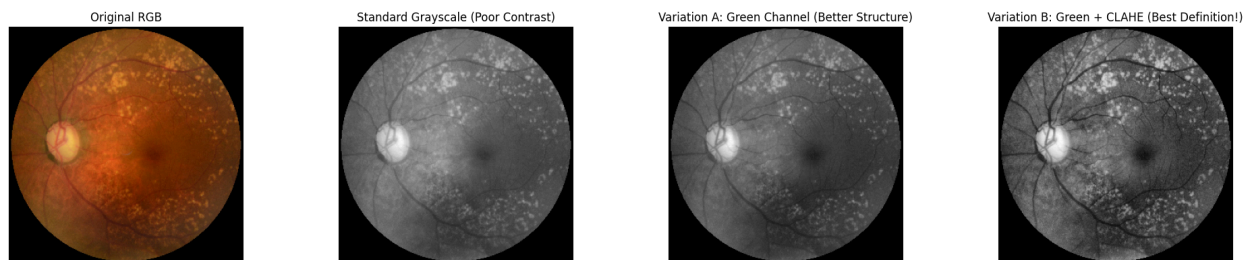


Figure 3: Preprocessing for the ARMD class.

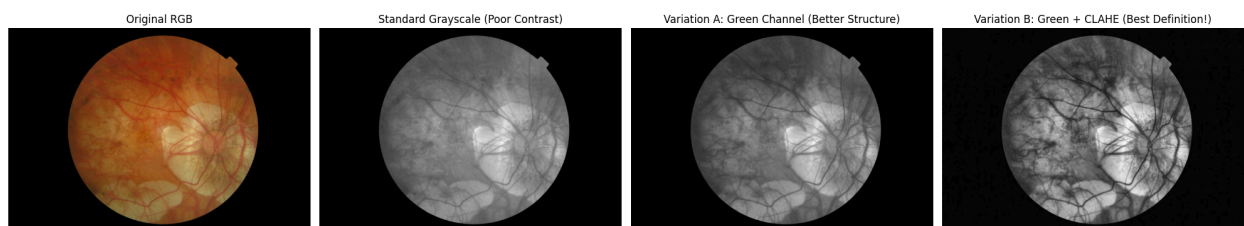


Figure 4: Preprocessing for the DR class.

4.1.2 Edge Detection

For this step, the results were not as expected, since the Canny method gave unreliable results for all thresholds and in all classes. For the Strict one, it removed a lot of the information, but the

worst was in the Sensitive, where the noise was covering the image even after applying denoising. As a result, Sobel gave the best results that can show most of the blood vessels. LoG method was even worse, it removed all the details as expected.

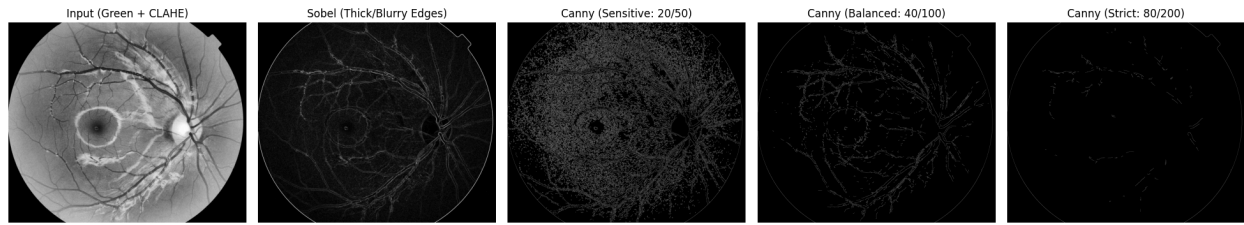


Figure 5: Edge Detection for the normal class.

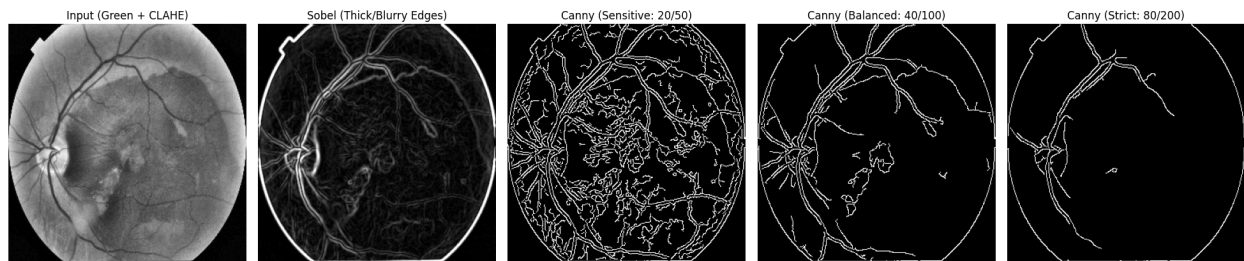


Figure 6: Edge Detection for the ARMD class.

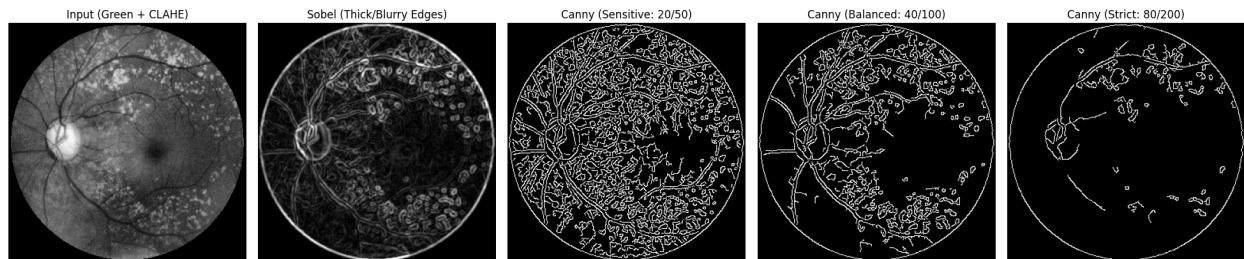


Figure 7: Edge Detection for the DR class.

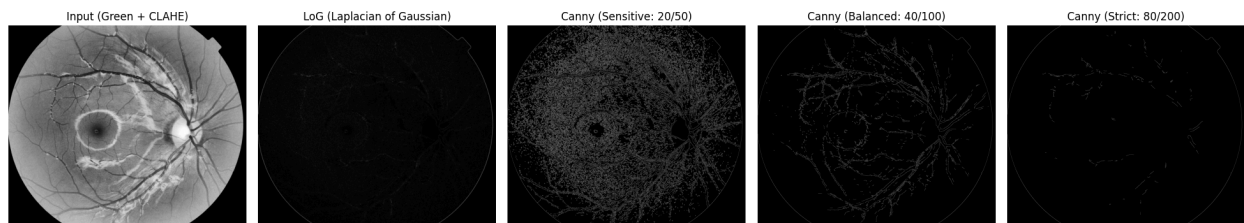


Figure 8: Edge Detection with LoG.

4.1.3 Corners Detection

In this detection, the results were mixed. Due to the changes in the images for normal, ARMD, and Diabetic retinopathy, the Shi-Tomasi method was more precise in finding where the blood vessels spread out in the normal images class, but in the ARMD images it gave messy results due to the hemorrhages and the appearance of drusen, all make it harder for corner detection. The same concept in Diabetic retinopathy since it leads to the neovascularization (the messy new veins)

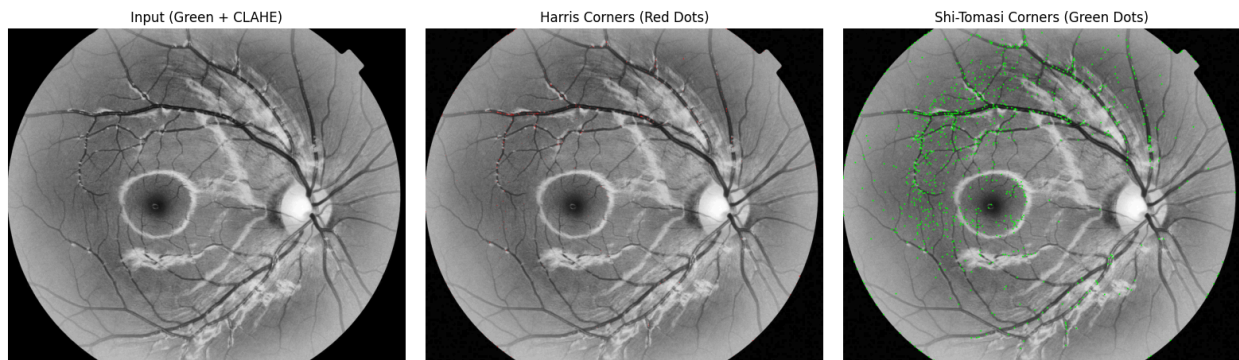


Figure 9: Corners Detection for the normal class.

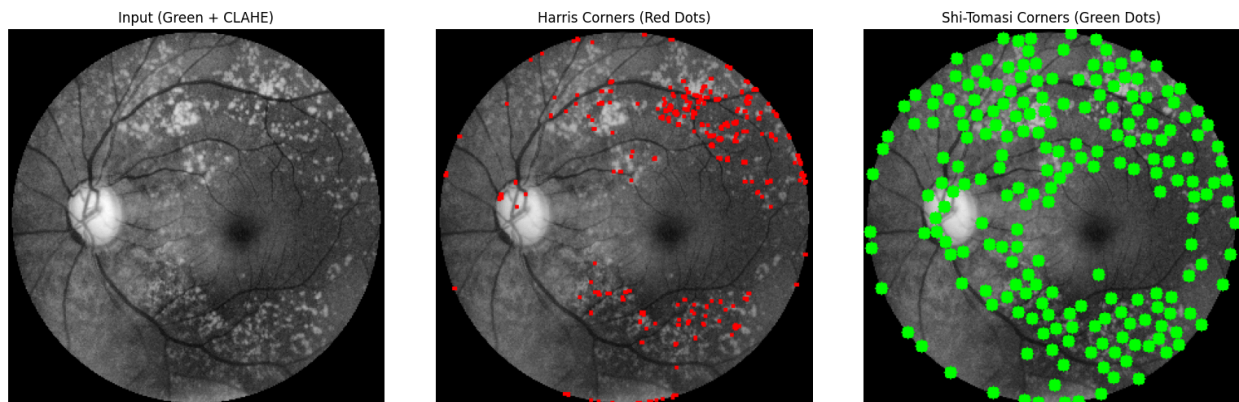


Figure 10: Corners Detection for the ARMD class.

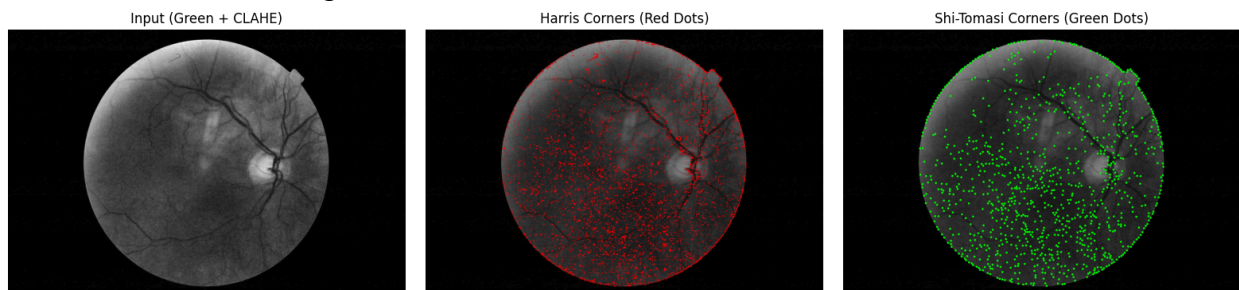


Figure 11: Corners Detection for the DR class.

4.1.4 Blobs Detection

In this step, the results are better, since the detected blobs actually refer to medically meaningful features. As mentioned above, the blob detection is meant to detect lesions. For the normal images, there was not much to detect since they are healthy and do not suffer from bright lesions. But for the ARMD images, SIFT managed to highlight the small drusen that are the main markers for ARMD patients. For the diabetic, it was even better since it was able to detect almost all the cotton wool spots, which are one of the important features for diabetes detection. In both ARMD and diabetes, ORB gave very messy results due to its algorithm, which was designed for faster, less accurate results than SIFT.

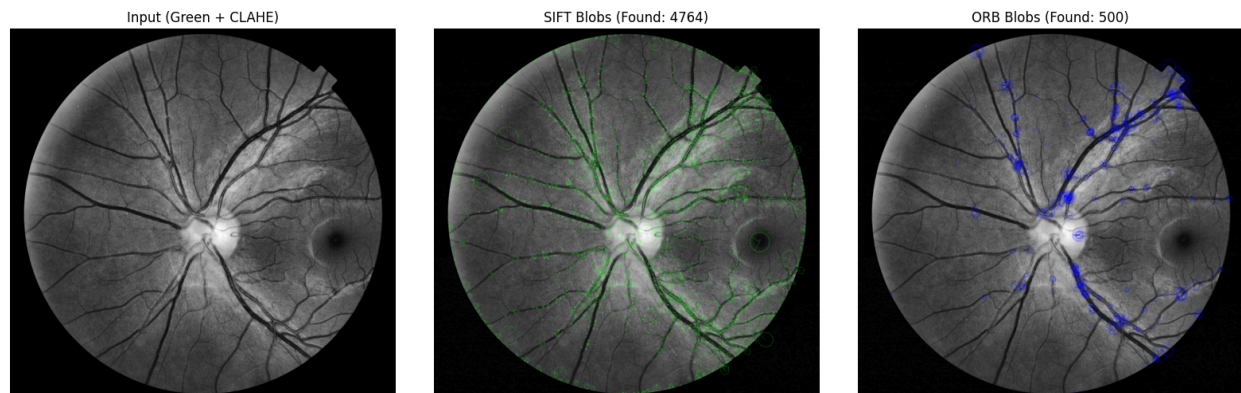


Figure 12: Blob Detection normal class.

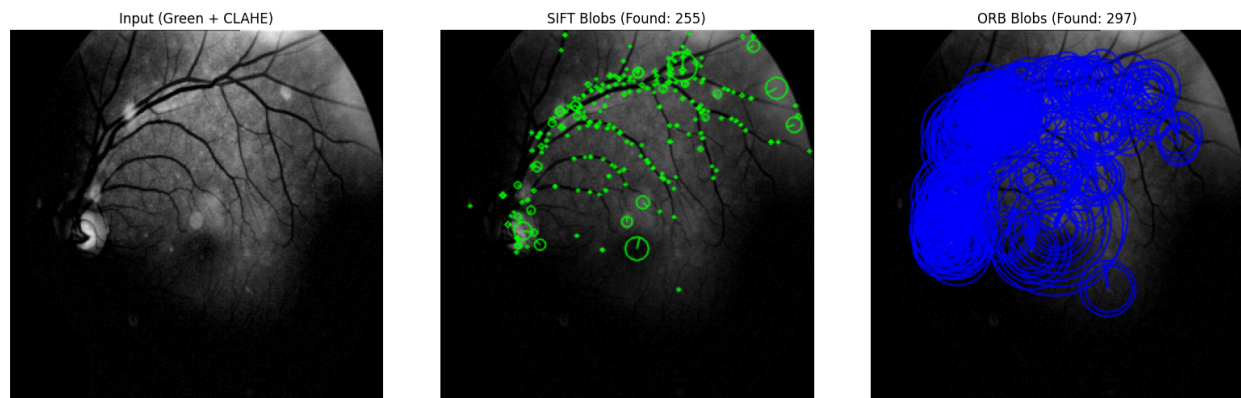


Figure 13: Blob Detection ARMD class.

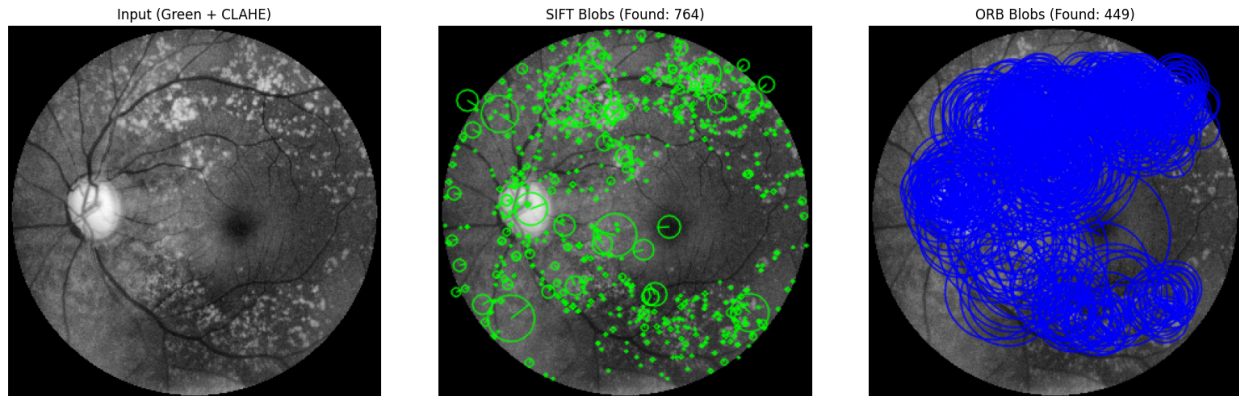


Figure 13: Blob Detection DR class.

4.2 Failure Cases

4.2.1 Canny Edge Detection

The canny method gave a systematic failure in the edge detection for the retinal fundus images for all the 3 thresholds. This is due to the logic Canny works with, since the retinal fundus images are very complex networks of blood vessels. Canny allows all small edges to be counted in the sensitive filter, which leads to allowing big number of noise to show. For the other thresholds, Canny tried to allow only bigger and more noticeable edges, but the problem that led to not showing most of the vessels is that the vessels are not big enough to notice a change with bigger vessels.

4.2.2 LoG Edg Detection

This algorithm gave very bad results due to its concept of calculating the edge. Other algorithms, like Canny and Sobel, use the first derivative for the mathematical equation when facing an edge (change of color density), which registers a single, clean line to make a border. However, LoG takes the second derivative, so it basically allows for a very sharp border (change in color), and this is not the case in the blood vessels.

4.2.3 ORB Blob Detection

Again, the problem is that the logic and algorithm of ORB are calculated based on ultra-fast detection that works similarly to SIFT but much faster, since it does not calculate the scale-space pyramid. It can locate the spot fast, but it always fails to give an accurate size. This is also due to the data it tried to detect, since most of the retinal images size is bright and have a lot of bright spots for ARMD and diabetes cases.

4.3 Quantitative Proxy Evaluation:

Table 1 (Sample Outputs)

Image ID	Condition	SIFT	ORB	Pathology
`1ffa9657...JPG`	Healthy	114	82	None
`1ffa9654...JPG`	Healthy	131	90	None
`0_aria_a_43...png`	ARMD	345	198	Moderate drusen
`0_ORID19...png`	ARMD	412	225	Severe drusen
`IMAGE_00361.jpg`	DR	590	410	Exudates
`IMAGE_01927.jpg`	DR	724	485	Severe lesions

Table 2 (Average Blob Counts)

Condition	AVG. SIFT Blobs	AVG. ORB Blobs	Interpretation
Normal	120	85	Baseline structural features
ARMD	380	210	Increased drusen presence
Diabetic Retinopathy	650	440	High density of exudates and hemorrhages

